

Constrained optimization

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Microeconomics 2

March 24, 2023

Modern Theory of Constrained Optimization

- Contributions by Kuhn-Tucker (KT), Arrow-Hurwitz-Uzawa (AHU), Slater, Arrow-Enthoven (AE), Karush.

Problem:

$$(P) \left\{ \begin{array}{ll} \max_{x \in X} & f(x_1, \dots, x_n) \\ \text{s.t.} & g^1(x_1, \dots, x_n) \geq 0 \\ & \cdot \\ & \cdot \\ & \cdot \\ & g^m(x_1, \dots, x_n) \geq 0 \\ \text{and} & (x_1, \dots, x_n) \in X \end{array} \right.$$

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^m \lambda_j g^j(x)$$

KT-I

- X open, convex
- f, g^j cont. diff. on X :
 - (i) $\nabla_i f(\hat{x}) + \sum_{j=1}^m \hat{\lambda}_j \nabla_i g^j(\hat{x}) = 0, i = 1, \dots, n$
 - (ii) $g^j(\hat{x}) \geq 0, j = 1, \dots, m$
 - (ii') $\hat{\lambda}_j g^j(\hat{x}) = 0, j = 1, \dots, m$
 - Conditions (ii') are called complementary slackness conditions.

KT-II

- $X = \mathbb{R}_+^n$
- f, g^j cont. diff. on $\mathbb{R}^n$

$$(i) \quad \nabla_i f(\hat{x}) + \sum_{j=1}^m \hat{\lambda}_j \nabla_i g^j(\hat{x}) \leq 0, \quad i = 1, \dots, n$$

$$(i') \quad \hat{x}_i [\nabla_i f(\hat{x}) + \sum_{j=1}^m \hat{\lambda}_j \nabla_i g^j(\hat{x})] = 0, \quad i = 1, \dots, n$$

$$(ii) \quad g^j(\hat{x}) \geq 0, \quad j = 1, \dots, m$$

$$(ii') \quad \hat{\lambda}_j g^j(\hat{x}) = 0, \quad j = 1, \dots, m$$

- Conditions (i') and (ii') are complementary slackness conditions

Sufficiency theorems

A-E sufficiency theorem 1:

Suppose f concave, g^j are quasiconcave on R_+^n .

Suppose $(\hat{x}, \hat{\lambda})$ satisfies KT-II.

Then $\hat{x}$ solves (P) .

A-E sufficiency theorem 2:

Suppose f, g^j are quasiconcave on $\mathbb{R}_+^n$.

Suppose $(\hat{x}, \hat{\lambda})$ satisfies KT-II.

If $\nabla f(\hat{x}) \neq 0$

then $\hat{x}$ solves (P) .

Two routes to solve (P) :

Sufficiency route: $KT \implies \text{global max}$

KT-I: f, g^j concave, X convex set

KT-II:

- AE1: f concave, g_j quasiconcave
- AE2: f, g^j quasiconcave, $\nabla f(\hat{x}) \neq 0$

Necessity route:

Step 1: Weierstrass $\implies \exists \hat{x}$ that solves (P)

Step 2: Necessary conditions:

If AHU C.Q.: KT-I.

If g^j quasiconcave, Slater's condition is satisfied, and $\nabla g^j(\hat{x}) \neq 0 \forall j = 1, \dots, m$: KT-II

Constraint Qualification

AHU Constraint Qualification (used with KT-I)

General qualification:

Let $E = \{j \in \{1, \dots, m\} : g^j(\hat{x}) = 0\}$

- 1 $E \neq \emptyset$
- 2 X is convex and $g^j(x)$ are convex functions $\forall j \in E$
- 3 There exists $h \in \mathbb{R}^n$ such that $h \nabla g^j(\hat{x}) > 0$ for all $j \in E$.

Simplified qualification:

- 1 X is a convex set and $g^j(x)$ is a convex function $\forall j = 1, \dots, m$
- 2 X is a convex set and $g^j(x)$ is a concave function $\forall j = 1, \dots, m$ and Slater's condition is satisfied

Slater's condition: $\exists \bar{x} \in X$ such that $g^j(\bar{x}) < 0 \forall j = 1, \dots, m$

Necessity theorem

A-E necessity theorem: (Global max $\implies$ KT)

Suppose $\hat{x}$ solves (P) .

Suppose g^j is quasiconcave $\forall j = 1, \dots, m$.

Assume Slater's condition is satisfied and $\nabla g^j(\hat{x}) \neq 0 \forall j = 1, \dots, m$.

Then $\exists \hat{\lambda}$ s.t. $(\hat{x}, \hat{\lambda})$ satisfies K-T.

Example:

- Let a, b, k be positive real numbers.
- Let $f : \mathbb{R} \Rightarrow \mathbb{R}$ be a concave, twice continuously differentiable function with $f'(x) > 0$ and $f''(x) < 0 \forall x \in \mathbb{R}$.
- Define: $\theta = f'(0)$

Solve the problem:

$$(P) = \begin{cases} \max f(x_1) + f(x_2) \\ \text{s.t. } ax_1 + bx_2 \leq k \\ \text{and } (x_1, x_2) \in \mathbb{R}_+^2 \end{cases}$$

Which route?

- The objective function is $F(x) = f(x_1) + f(x_2)$.
- $\nabla f(x) = \begin{bmatrix} f'(x_1) \\ f'(x_2) \end{bmatrix}$ $H_f(x) = \begin{bmatrix} f''(x_1) & 0 \\ 0 & f''(x_2) \end{bmatrix}$
 $f''(x_1) < 0$, $f''(x_1)f''(x_2) - 00 > 0$, so f is concave, thus F is concave.
- The constraint can be rewritten as $g(x) = k - ax_1 - bx_2 \geq 0$. It is linear, so concave, which implies quasiconcave.
- So can use sufficiency route
- Note: $X = \mathbb{R}_+^2$, so we use KT-II.
- $\mathcal{L} = f(x_1) + f(x_2) + \lambda[k - ax_1 - bx_2]$

KT-II conditions:

$$(ia) \quad f'(x_1^*) + \lambda(-a) \leq 0$$

$$(ib) \quad f'(x_2^*) + \lambda(-b) \leq 0$$

$$(ia') \quad x_1^*[f'(x_1^*) - \lambda a] = 0$$

$$(ib') \quad x_2^*[f'(x_2^*) - \lambda b] = 0$$

$$(ii) \quad k - ax_1^* - bx_2^* \geq 0$$

$$(ii') \quad \lambda^*[k - ax_1^* - bx_2^*] = 0$$

Go through cases using complimentary slackness conditions (ia'), (ib'), and (ii'). This gives parameter restrictions for corner vs. interior solution.